Literature-Implied Partial Answer: Weak-FPP Under A Block-Basis Hypothesis

FA/Banach run `fa_banach_001`

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Status

This packet records a *literature-implied partial subcase*, not an original proof and not a full solution of the unconditional-basis weak-FPP problem. Barroso arXiv:2012.10849 states that it remains open whether every Banach space with an unconditional basis has the weak fixed point property. Barroso's later paper arXiv:2302.11287 proves an affirmative result for a structurally restricted subcase: bases whose block basis sequences always contain a subsequence equivalent to the original basis.

Original Question

The original source is:

> C. S. Barroso, *Isometric embeddings of Banach spaces under optimal projection constants*, arXiv:2012.10849.

In the metric fixed point application section, after recalling Lin's theorem, the paper says that it is still open whether every Banach space with an unconditional basis has the weak-FPP.

In the sequel, we will show how this result can be applied in Metric Fixed Point Theory. One of the major concerns in this research area concerns the weak fixed point property (weak-FPP, for short). Let C be a weakly compact convex subset of a Banach space X. Does every nonexpansive (i.e., 1-Lipschitz) mapping $T\colon C \to C$ have a fixed point? When this happen, we say that C has the weak-FPP. Further, we say that X has the weak-FPP if every weakly compact convex subset of X has the weak-FPP.

In 1985, P.-K. Lin [Lin] proved the following beautiful result:

Theorem III. Let X be Banach space with a $\mathcal{D}$-unconditional basis satisfying $\mathcal{D} < (\sqrt{33}-3)/5$. Then X has the weak fixed point property.

Since then, significant attention has been paid to the weak-FPP in Banach spaces with Schauder basis. In particular, it is still an open question as to whether or not every Banach space with an unconditional basis has the weak-FPP. For this setting, the strategy of transferring fixed point analysis to the framework of [X] (the ultrapower of X) has always been a fruitful strategy for studying fixed point property in spaces with Schauder basis. As a consequence, if some quantities related to basis projections can become conveniently small, then Lin's approach [Lin] can be successfully applied to obtain better results. This is the key idea behind our second result (Theorem B). Note

The same source paper proves a shrinking D-unconditional subcase with $D < \sqrt{6}-1$. That same-paper result is treated only as context here, not as a packet-worthy literature answer.

Supporting Literature

The supporting source is:

> C. S. Barroso, *Remarks on the FPP in Banach spaces with unconditional Schauder basis*, arXiv:2302.11287.

The later paper explicitly formulates the broad unconditional-basis question and the 1-suppression unconditional version:

be relaxed. Precisely:

($\mathcal{Q}1$) Does every X with an unconditional basis have the weak-FPP?

($\mathcal{Q}2$) Does every X with a 1-suppression unconditional basis have the weak-FPP?

One technical difficulty behind these issues is that subsequences of weakly null sequences can have large unconditional constants. Despite that, Lin [39] showed that under super-reflexivity ($\mathcal{Q}2$) has an affirmative answer. However, Lin's approach does not seem to be effective in general spaces with such bases. In line with these matters, the author [7] gave a small step towards ($\mathcal{Q}1$) by proving the following improvement: Every Banach space with a shrinking $\mathcal{D}$-unconditional basis such that $\mathcal{D} < \sqrt{6}-1$ has the weak-FPP. Another

Its Theorem 3.11 states:

This contradiction proves the theorem. □

Theorem 3.11. *Let X be a Banach space with a Schauder basis (e_i). Assume that every block basis sequence has a subsequence equivalent to the basis. Then X has the weak fixed point property in each of the following two situations:*

(i) *(e_i) is a 1-suppression unconditional basis.*

(ii) *(e_i) is a 1-spreading basis.*

Proof Intuition

The later theorem supplies the missing fixed point conclusion directly once the original question is narrowed by a strong block-basis regularity hypothesis. The branch in which the given basis is 1-suppression unconditional is a subcase of having an unconditional basis. The additional assumption, that every block basis sequence contains a subsequence equivalent to the original basis, is the structural ingredient that lets the supporting paper run its spreading-model and weakly-null sequence argument.

Theorem Recorded

Claim. Let X be a Banach space with a Schauder basis (e_i). Suppose:

1. (e_i) is 1-suppression unconditional; and
2. every block basis sequence of (e_i) has a subsequence equivalent to (e_i).

Then X has the weak fixed point property.

Proof

This is exactly the 1-suppression unconditional branch of Theorem 3.11 in Barroso arXiv:2302.11287. The theorem assumes that X has a Schauder basis (e_i) such that every block basis sequence has a subsequence equivalent to the basis. It then concludes that X has the weak fixed point property in each of two cases, the first being that (e_i) is a 1-suppression unconditional basis. These are precisely the two hypotheses listed above, so the conclusion follows.

Because a 1-suppression unconditional basis is unconditional, this gives an affirmative answer to the original unconditional-basis question for this restricted class of spaces.

Scope And Limitations

- The full question "Does every Banach space with an unconditional basis have the weak-FPP?" remains open in the sources checked here.
- The packet does not answer the question for all 1-suppression unconditional bases; the block-basis-equivalence hypothesis is part of the result.
- Theorem 3.11 also covers a 1-spreading-basis branch. This packet does not use that branch as the main match, because the direct subcase of the original unconditional-basis problem is the 1-suppression unconditional branch.

- The packet records a direct literature implication from arXiv:2302.11287; it is not a new proof from this run.

Verification Notes

The source statement was checked in arXiv:2012.10849, page 5. The supporting theorem was checked in arXiv:2302.11287, page 8. A bounded web search on 2026-06-14 for exact phrases around "every Banach space with an unconditional basis" and "weak fixed point property" found the same 2023 follow-up paper and did not show a later full resolution of the broad question. The later paper itself phrases the broad questions ($Q1$) and ($Q2$), cites the 2020 paper as a small step, and provides the restricted Theorem 3.11 recorded above.

No computational verification is relevant.

Human Review Recommendation

Keep the classification as *literature-implied partial subcase*. The main reviewer check is that the block-basis assumption in Theorem 3.11 has not been accidentally omitted when comparing to the original open question.

References

[1] C. S. Barroso, *Isometric embeddings of Banach spaces under optimal projection constants*, arXiv:2012.10849.

[2] C. S. Barroso, *Remarks on the FPP in Banach spaces with unconditional Schauder basis*, arXiv:2302.11287.